AAC - BIOcean5D Coral Microbiomes: 03 - Phylogenetic Tree

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Session Information **14**

Purpose of this file This file contains all of the code for producing a curated OTU table from the 16S microbial sequence data for the Armstrong et al. manuscript entitled **"XXX"**, published by XXX.

If you have any questions, you are welcome to email the corresponding author at eja98@cornell.edu or tweet him at EricJ_Armstrong. The goal is to create a phylogenetic tree

1. Load in preprocessed phyloseq object.
2. Create ASV fasta file from the phyloseq object.
3. Align the 16S sequences from fasta file with MAFFT.
4. Create a tree with FastTree2.

Finally, write data file of phyloseq output, which will be stored in Data/03-AAC-PhylogeneticTree/.

If you have any questions, you are welcome to email the corresponding author at eja98@cornell.edu or tweet him at EricJ_Armstrong.

1 - Prepare Workspace

Load necessary R packages and set default parameters

1.1 - Define Paths

```
# a - Define the main project directory ----
  dir_proj <- "C:/Users/uax75/OneDrive/Documents/R/AAC-BIOcean5D-Coral_Microbiome/"

# b - Define the directory containing the input data ----
  dir_in <- "C:/Users/uax75/OneDrive/Documents/R/AAC-BIOcean5D-Coral_Microbiome/Data/"

# c - Define the directories that will contain output Files/Figures/Tables ----
  dir_out_dat <-
    "C:/Users/uax75/OneDrive/Documents/R/AAC-BIOcean5D-Coral_Microbiome/Data/03-AAC-PhylogeneticTree/"
  dir.create(dir_out_dat, recursive = TRUE)
  dir_out_fig <-
    "C:/Users/uax75/OneDrive/Documents/R/AAC-BIOcean5D-Coral_Microbiome/Figures/03-AAC-PhylogeneticTree
  dir.create(dir_out_fig, recursive = TRUE)
  dir_out_tab <-
    "C:/Users/uax75/OneDrive/Documents/R/AAC-BIOcean5D-Coral_Microbiome/Tables/03-AAC-PhylogeneticTree/
  dir.create(dir_out_tab, recursive = TRUE)
```

1.2 - Set Defaults

```
# a - Define the output prefix for generated files ----
  pref_out <- "03-AAC-PhylogeneticTree_"

# b - Set the random seed value ----
  set.seed(123)
```

1.3 - Load Libraries

```
# a - load necessary packages ----
  pacman::p_load(tidyverse, phyloseq, ggtree, phytools, ggplot2, expss, ape,
                 install = FALSE)

  # DECIPHER and Biostrings are deliberately NOT attached. Attaching either
  # pulls in S4Vectors and BiocGenerics, whose exports (rename, select,
  # setdiff, intersect, union, ...) mask the dplyr verbs this script relies on
  # - attaching them after tidyverse sent `rename(RefSeq = 2)` below to
  # S4Vectors::rename and halted the knit. Both are called with the `::`
  # prefix in sections 4b.1 and 4b.2, which needs them installed but not
  # attached. Check availability early so the failure is clear.
  stopifnot(requireNamespace("DECIPHER",   quietly = TRUE),
            requireNamespace("Biostrings", quietly = TRUE))
```

1.4 - Load Phyloseq Object

```
# a - Load physeq ----
  load(paste(dir_in,
             "02-AAC-PreProcessing/",
             "02-AAC-PreProcessing_physeq_noControlsChlorosMitos.RData",
             sep = ""))
```

2 - Create Fasta File of ASV and their Sequences

This fasta file will be used to create our alignment in MAFFT

```
# a - pull out ASV seqs and ASV names ----
  df_asv_seq <-
    physeq_noControlsChlorosMitos@refseq %>%
    data.frame() %>%
    rownames_to_column(var = "ASV") %>%
    dplyr::rename(RefSeq = 2)

# b - Add the > to make fasta header ----
  df_asv_seq$ASV <- paste0(">",df_asv_seq$ASV)
  #View(asv_seq_df)

# c - Create the fasta object ----
  fasta_asv_seq <- c(rbind(df_asv_seq$ASV, df_asv_seq$RefSeq))
  head(fasta_asv_seq)
#> [1] ">ASV1"
#> [2] "TACGGAGGGTGCGAGCGTTAATCGGAATTACTGGGCGTAAAGCGTGCGTAGGCGGCTTTTTAAGTTGGATGTGAAAGCCCCGGGCTCAACCTGGG
#> [3] ">ASV2"
#> [4] "TACGGAGGATGCAAGCGTTATCCGGAATTATTGGGCGTAAAGCGTCCGCAGGCGGCACTATAAGTCTGCTGTCAAAGACCGCAGCTCAACTGCGG
#> [5] ">ASV4"
#> [6] "TACGGGAGTGGCAAGCGTTATCCGGAATTATTGGGCGTAAAGCGTCCGCAGGCGGCCCTTCAAGTCTGCTGTTAAAAAGTGGAGCTTAACTCCAT

# d - Write to a file ----
  write(fasta_asv_seq,
```

```
        file = paste(dir_out_dat, pref_out,
                     "PhyseqASVs.fasta",
                     sep = ""))
```

3 - Align the 16S sequences from fasta file with MAFFT

```
# Write bash code to run mafft
# First provide the path to mafft
export PATH=/programs/mafft/bin:$PATH
# change directories to provide the fasta file we made above
cd /workdir/eja98/BIO5D_Data/Data/03-AAC-PhylogeneticTree/
pwd

# Set a seed  - using same seed as before for consistency
RANDOM=238428

# Run Mafft
# To test in the shell directly from Rmd
# mac: command + option + enter
# Windows: control + alt + enter
# For now, use default options, note the version
# MAFFT automatically knows that it's a nucleotide alignment
/programs/mafft/bin/mafft --auto 03-AAC-PhylogeneticTree_PhyseqASVs.fasta > 03-AAC-PhylogeneticTree_MAFF
```

4 - FastTree2

```
# Provide export path to fasttree
export PATH=/programs/FastTree-2.1.11:$PATH

# cd into the alignment file folder
cd /workdir/eja98/BIO5D_Data/Data/03-AAC-PhylogeneticTree/
pwd


FastTree -nt -gtr -fastest -log FastTree.log 03-AAC-PhylogeneticTree_MAFFT_aligned_PhyseqASVs.fasta > 0
```

4b - Alignment and tree, run locally in R

The two chunks above are the commands that were run on the Cornell BioHPC node. They are kept for the record and are not evaluated. That cluster is no longer available, so the alignment and the tree are now produced from this script.

- **Alignment** - `DECIPHER::AlignSeqs()` replaces `mafft --auto`. It performs the same kind of iterative profile alignment, runs inside R, and handles a set of this size without difficulty.

- **Tree** - FastTree2 is still used, with the identical flags (`-nt -gtr -fastest`), called through the Windows binary rather than a cluster module. The tree method is therefore unchanged from the original analysis. Download `FastTree.exe` once from http://www.microbesonline.org/fasttree/ and either place it on the PATH or point `FASTTREE_EXE` at it.

Both steps cache to disk and are skipped when their output already exists, so re-knitting this document does not repeat hours of work. Delete the output file to force a rebuild.

```
# a - Paths ----
  fa_in   <- paste0(dir_out_dat, pref_out, "PhyseqASVs.fasta")
  fa_algn <- paste0(dir_out_dat, pref_out, "DECIPHER_aligned_PhyseqASVs.fasta")

# b - Align, or reuse an existing alignment ----
  # Reuse an existing alignment ONLY if it covers exactly the current ASV set.
  # A bare file.exists() check is unsafe here: the repository still carries
  # alignment and tree files from the May 2026 run, built before the amplicon
  # window was repaired, and silently reusing those would mismatch the taxa.
  reuse_algn <- FALSE
  if (file.exists(fa_algn)) {
    old <- try(Biostrings::readDNAStringSet(fa_algn), silent = TRUE)
    reuse_algn <- !inherits(old, "try-error") &&
      setequal(names(old), taxa_names(physeq_noControlsChlorosMitos))
    if (!reuse_algn)
      message("Existing alignment does not match the current ASV set",
              " - realigning.")
  }

  if (!reuse_algn) {
    dna <- Biostrings::readDNAStringSet(fa_in)
    message("Aligning ", length(dna), " ASVs with DECIPHER::AlignSeqs ...")
    algn <- DECIPHER::AlignSeqs(dna,
                                anchor     = NA,
                                processors = NULL,   # use all cores
                                verbose    = TRUE)
    Biostrings::writeXStringSet(algn, filepath = fa_algn)
  } else {
    algn <- Biostrings::readDNAStringSet(fa_algn)
    message("Reusing existing alignment (", length(algn), " sequences)")
  }

  algn
#> DNAStringSet object of length 13673:
#>         width seq                                               names
#>     [1]   740 TACGGAGGGTGCG--------...CACTCTGGCTTGAC------- ASV1
#>     [2]   740 TACGGAGGATGCA--------...TCTACTGGGCCAT-------- ASV2
#>     [3]   740 TACGGGAGTGGCA--------...TCTGCTGGGCCAT-------- ASV4
#>     [4]   740 TACGGAGGGTGCG--------...CTCCCTGGCCAAA-------- ASV5
#>     [5]   740 TACGGGAGTGGCA--------...TCTGCTGGGCCAT-------- ASV6
#>     ...   ... ...
#> [13669]   740 TACGTAGGATGCA--------...CTTGCTGGGGCAGT------- ASV14413
#> [13670]   740 TACGGAGGGGGCT--------...CTACCTGGACCGGT------- ASV14414
#> [13671]   740 TACGTAGGGTGCA--------...TCCCCTGGGCCTG-------- ASV14416
#> [13672]   740 TACAGAGGGTGCA--------...CTCTCTGGGCTTAC------- ASV14417
#> [13673]   740 TACGGAGGGTGCA--------...CTCGCTAAACTAT-------- ASV14418
```

```
# a - Locate the FastTree executable ----
  # Set this once per machine, e.g. in .Rprofile:
  #   Sys.setenv(FASTTREE_EXE = "C:/bin/FastTree.exe")
  fasttree_exe <- Sys.getenv("FASTTREE_EXE", unset = "FastTree.exe")

  tre_out <- paste0(dir_out_dat, pref_out, "PhyseqASVs_Unrooted.tree")
  tre_log <- paste0(dir_out_dat, "FastTree.log")

# b - Build the tree, or reuse an existing one ----
  # Same caution as for the alignment: the May 2026 tree sits at exactly this
  # path, so validate the tip labels rather than trusting the filename.
  reuse_tree <- FALSE
  if (file.exists(tre_out)) {
    old_tree <- try(ape::read.tree(tre_out), silent = TRUE)
    reuse_tree <- !inherits(old_tree, "try-error") &&
      setequal(old_tree$tip.label, taxa_names(physeq_noControlsChlorosMitos))
    if (!reuse_tree)
      message("Existing tree does not match the current ASV set - rebuilding.")
  }

  if (!reuse_tree) {

    if (Sys.which(fasttree_exe) == "" && !file.exists(fasttree_exe))
      stop("FastTree executable not found: '", fasttree_exe, "'.\n",
           "Download FastTree.exe from http://www.microbesonline.org/fasttree/ ",
           "and either place it on the PATH or set FASTTREE_EXE to its full ",
           "path, e.g. Sys.setenv(FASTTREE_EXE = 'C:/bin/FastTree.exe').")

    message("Running FastTree (-nt -gtr -fastest) on ", basename(fa_algn), " ...")
    status <- system2(fasttree_exe,
                      args   = c("-nt", "-gtr", "-fastest",
                                 "-log", shQuote(tre_log, type = "cmd"),
                                 shQuote(fa_algn, type = "cmd")),
                      stdout = tre_out)

    if (!identical(as.integer(status), 0L) || !file.exists(tre_out))
      stop("FastTree failed with status ", status)

    message("Tree written to ", tre_out)
  } else {
    message("Reusing existing tree: ", tre_out)
  }
```

5 - Tree Visualizations

5.1 - Load in the tree —-

```
unrooted_tree <- read.tree(
  paste(dir_in, "03-AAC-PhylogeneticTree/",
        "03-AAC-PhylogeneticTree_PhyseqASVs_Unrooted.tree", sep = ""))
```

```
unrooted_tree
#>
#> Phylogenetic tree with 13673 tips and 13671 internal nodes.
#>
#> Tip labels:
#>   ASV12690, ASV12758, ASV14401, ASV11762, ASV9844, ASV14305, ...
#> Node labels:
#>   , 0.850, 0.752, 0.878, 0.780, 0.111, ...
#>
#> Unrooted; includes branch length(s).
```

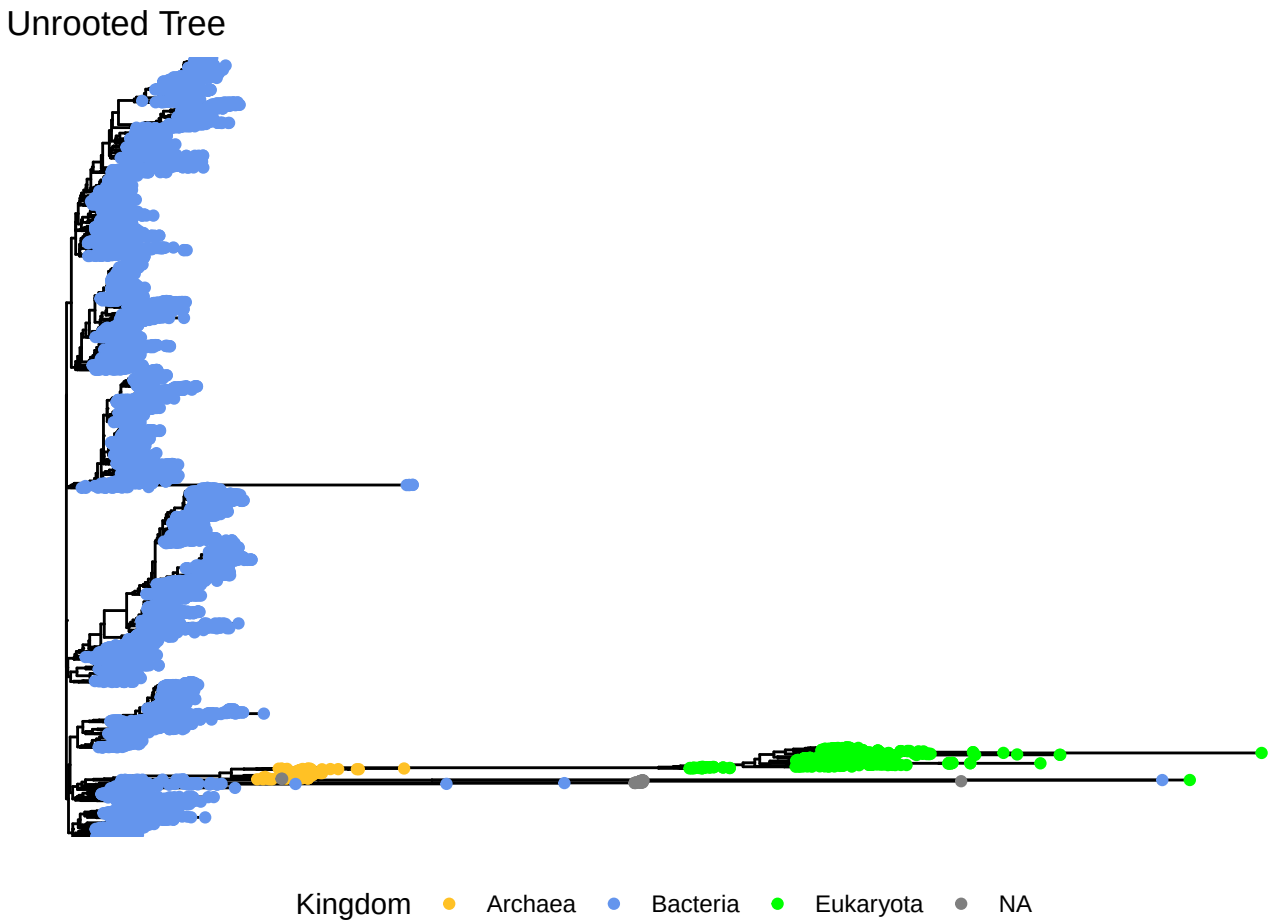
5.2 - Merge tree and phyloseq objects —-

```
# Intuition check
stopifnot(ntaxa(physeq_noControlsChlorosMitos) == ntaxa(unrooted_tree))

# Merge the tree with the phyloseq object
unrooted_physeq <-
  merge_phyloseq(physeq_noControlsChlorosMitos, unrooted_tree)
```

5.3 - Plot the tree —-

```
# Make a basic tree
kingdom_tree <-
  ggtree(unrooted_physeq) +
  # color tips by kingdom
  geom_tippoint(mapping = aes(color = Kingdom)) +
  scale_color_manual(
    values = c("goldenrod1", "cornflowerblue", "green", "red")) +
  # Add title
  labs(title = "Unrooted Tree") +
  #move the legend to the bottom
  theme(legend.position = "bottom")
kingdom_tree
```



5.4 - Prune bad taxa —-

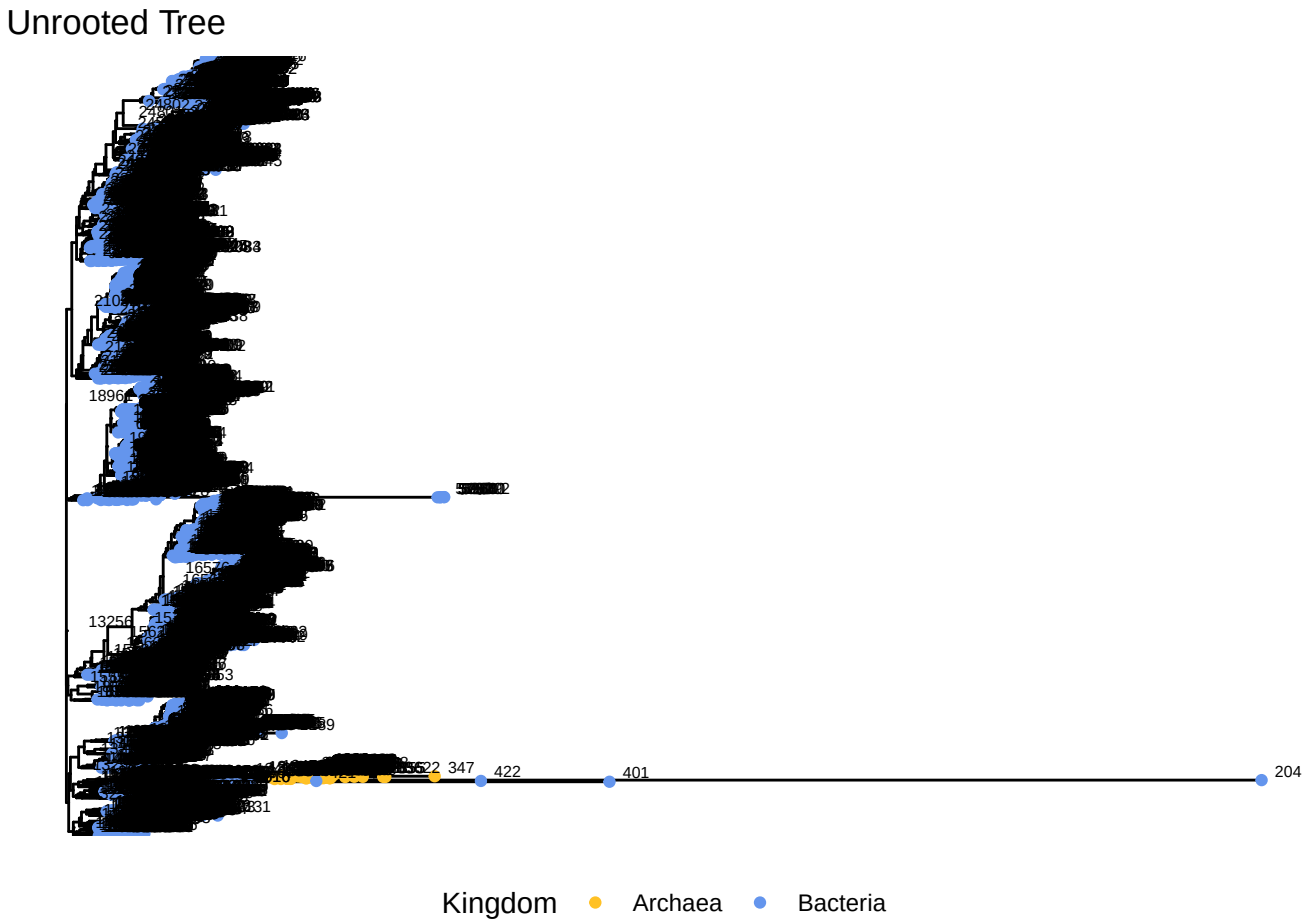
```
# Function from Joey McMurdie: https://github.com/joey711/phyloseq/issues/652
pop_taxa = function(physeq, badTaxa){
  allTaxa <-  taxa_names(physeq)
  allTaxa <- allTaxa[!(allTaxa %in% badTaxa)]
  return(prune_taxa(allTaxa, physeq))}

# Let's use the pop_taxa function :)
# Recreate a phyloseq object without eukaryotic taxa
EukTaxa <- unrooted_physeq %>%
  subset_taxa(., Kingdom %in% c("Eukaryota", NA)) %>%
  otu_table(.) %>%
  rownames(.)

unrooted_physeq_noEuk <-
  unrooted_physeq %>%
  pop_taxa(., EukTaxa)

kingdom_tree_noEuk <-
```

```
  ggtree(unrooted_physeq_noEuk) +
  # color tips by kingdom
  geom_tippoint(mapping = aes(color = Kingdom)) +
  scale_color_manual(values = c("goldenrod1", "cornflowerblue", "red")) +
  # Add the node label
  geom_text(aes(label=node), hjust= -0.5, vjust = -0.3, size = 2) +
  # Add title
  labs(title = "Unrooted Tree") +
  #move the legend to the bottom
  theme(legend.position = "bottom")
kingdom_tree_noEuk
```




```
# Let's use the pop_taxa function :)
# Recreate a phyloseq object without the long-branch-length taxa

# 1. Extract the tree from your phyloseq object (named 'ps')
tree <- phy_tree(unrooted_physeq_noEuk)

# 2. Find terminal edge (branch) lengths corresponding to each tip
is_tip <- tree$edge[, 2] <= Ntip(tree)
```

```
tip_nodes <- tree$edge[is_tip, 2]
terminal_lengths <- tree$edge.length[is_tip]

# 3. Create a named vector matching tip IDs (taxa names) to branch lengths
names(terminal_lengths) <- tree$tip.label[tip_nodes]

# 4. Set your threshold (e.g., branches longer than 0.37 or 89th percentile)
threshold <- 0.37
long_branch_ids <- names(terminal_lengths[terminal_lengths > threshold])

# View the retrieved taxon IDs
print(long_branch_ids)
#>  [1] "ASV11922" "ASV14121" "ASV13576" "ASV14412" "ASV14016" "ASV13972"
#>  [7] "ASV3504"  "ASV9890"  "ASV2596"  "ASV13770" "ASV13514" "ASV12010"
#> [13] "ASV14108" "ASV11880" "ASV14053" "ASV11933" "ASV12150" "ASV13298"
#> [19] "ASV13728"

# Extract the taxonomy row for that specific ASV
tax_info <- tax_table(unrooted_physeq_noEuk)[long_branch_ids, ]

# View as a named character vector or convert to data frame
print(tax_info)
#> Taxonomy Table:     [19 taxa by 10 taxonomic ranks]:
#>          Kingdom    Phylum            Class                 Order
#> ASV11922 "Bacteria" "Patescibacteria" "Parcubacteria"       NA
#> ASV14121 "Bacteria" NA                NA                    NA
#> ASV13576 "Bacteria" NA                NA                    NA
#> ASV14412 "Bacteria" "Bacteroidota"    "Bacteroidia"         "Cytophagales"
#> ASV14016 "Bacteria" "Pseudomonadota"  NA                    NA
#> ASV13972 "Archaea"  "Nanoarchaeota"   "Nanoarchaeia"        "Woesearchaeales"
#> ASV3504  "Bacteria" "Pseudomonadota"  NA                    NA
#> ASV9890  "Bacteria" NA                NA                    NA
#> ASV2596  "Bacteria" NA                NA                    NA
#> ASV13770 "Bacteria" "Pseudomonadota"  "Alphaproteobacteria" NA
#> ASV13514 "Bacteria" "Pseudomonadota"  "Alphaproteobacteria" "Rickettsiales"
#> ASV12010 "Bacteria" NA                NA                    NA
#> ASV14108 "Bacteria" "Spirochaetota"   "Brachyspirae"        "Brachyspirales"
#> ASV11880 "Bacteria" NA                NA                    NA
#> ASV14053 "Bacteria" "Cyanobacteriota" "Cyanobacteriia"      NA
#> ASV11933 "Bacteria" NA                NA                    NA
#> ASV12150 "Bacteria" NA                NA                    NA
#> ASV13298 "Bacteria" NA                NA                    NA
#> ASV13728 "Bacteria" "Pseudomonadota"  "Gammaproteobacteria" "Pseudomonadales"
#>          Family                Genus              Species Subspecies ASV
#> ASV11922 NA                    NA                 NA      NA         "ASV_11922"
#> ASV14121 NA                    NA                 NA      NA         "ASV_14121"
#> ASV13576 NA                    NA                 NA      NA         "ASV_13576"
#> ASV14412 "Cyclobacteriaceae"   "Roseivirga"       NA      NA         "ASV_14412"
#> ASV14016 NA                    NA                 NA      NA         "ASV_14016"
#> ASV13972 "Incertae Sedis"      "SCGC AAA286-E23" NA      NA         "ASV_13972"
#> ASV3504  NA                    NA                 NA      NA         "ASV_3504"
#> ASV9890  NA                    NA                 NA      NA         "ASV_9890"
#> ASV2596  NA                    NA                 NA      NA         "ASV_2596"
```

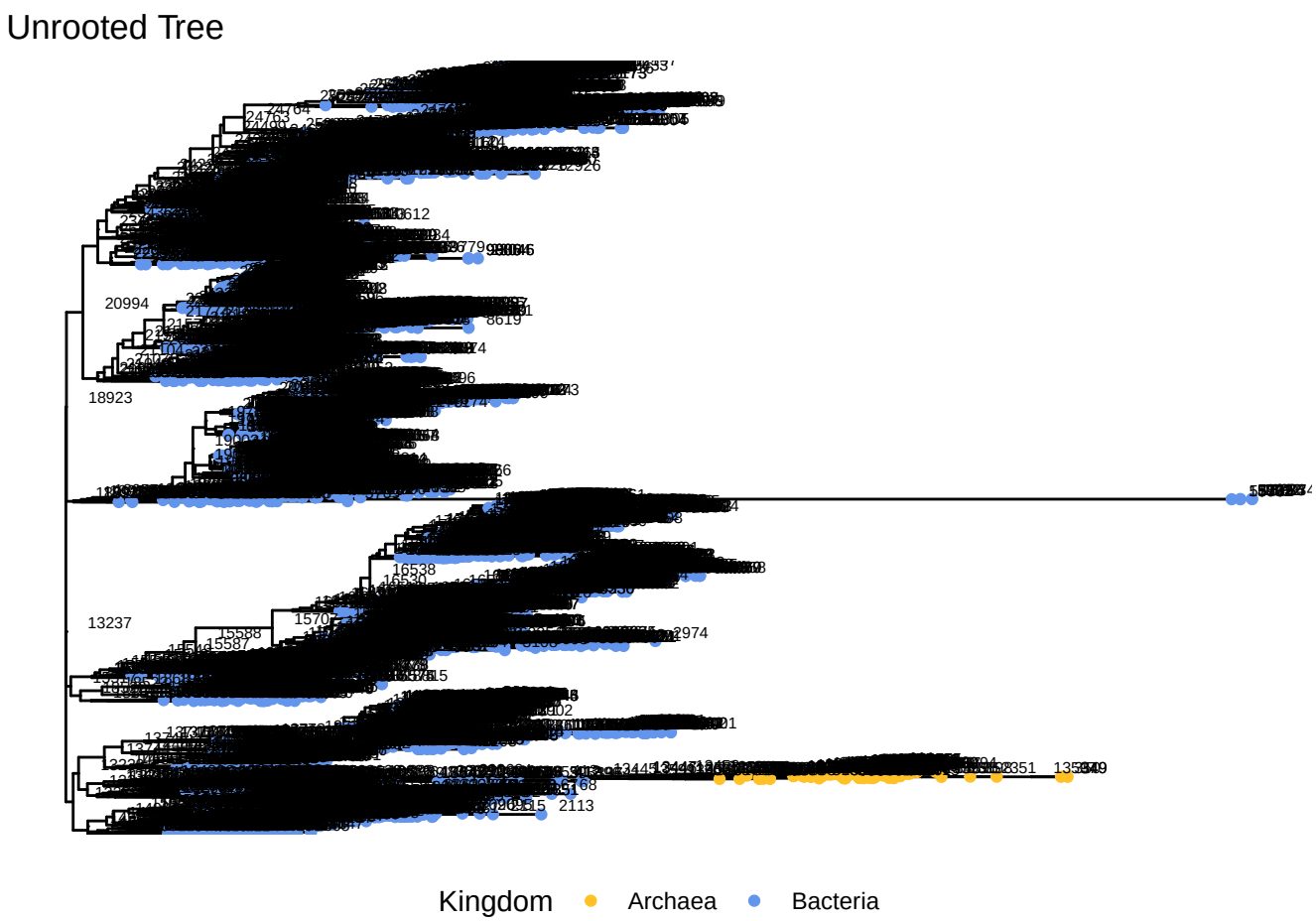
```
#> ASV13770 NA                     NA                NA      NA          "ASV_13770"
#> ASV13514 NA                     NA                NA      NA          "ASV_13514"
#> ASV12010 NA                     NA                NA      NA          "ASV_12010"
#> ASV14108 "Brachyspiraceae"      "Brachyspira"     NA      NA          "ASV_14108"
#> ASV11880 NA                     NA                NA      NA          "ASV_11880"
#> ASV14053 NA                     NA                NA      NA          "ASV_14053"
#> ASV11933 NA                     NA                NA      NA          "ASV_11933"
#> ASV12150 NA                     NA                NA      NA          "ASV_12150"
#> ASV13298 NA                     NA                NA      NA          "ASV_13298"
#> ASV13728 "Endozoicomonadaceae" "Endozoicomonas"  NA      NA          "ASV_13728"
#>          ASVseqs
#> ASV11922 "TACGGAGACCTCAAGCGTTATCCGGAATCATTGGGCGTAAAGCGTACCGATCGGTGGTTTGGTAAGTCTGTTGTTAAAGACCGAAGCTCA
#> ASV14121 "TACGTAGGTGGCAAGCGTTATTCGGAATTACTGGGCGTAAAGCGTTCGTAGGCGATTTTGTAGGTCCTTGATTAAATACTTCAGCTCAAC
#> ASV13576 "TACGTAGGGCGCGAGCGTTATCCGGATTTACTGGGCGTAAAGAGTCGTGTAGGTGGTTTGTTACATCATGCTTGAAATCCTATGGCTCAA
#> ASV14412 "AACGTAGGACGCAAGCGTTATCCGGAATTACTGGGCGTAAAGCGCGCGCAGGCGGGGTGTTAAGTCAGCCGTGAAAGCTCCCGGCTAAAC
#> ASV14016 "TACATGAGTGGTGAGCGTTGCTTTTAATTGTTAGGTGTAAAACATCCTGAGAAACAATTATGGGTTATATATGTTAGTATACTATAATTG
#> ASV13972 "TACCGGCGCCACAAGTGGAAATCACGATTATTGGGTCTAAAGTGTTCGTAGCTGGTTATATAAGTTTCTTGTTAAAGTTCTGTAGCTTAA
#> ASV3504  "CACGATTAACCCAAGTCAATAGAAGCCGGCGTAAAGAGTGTTTTAGATCACCCCCTCCCCAATAAAGCTAAAACTCACCTGAGTTGTAAA
#> ASV9890  "GACGGGGGATGCGAGCGTTATCCGGAATGACTGGGCGTAAAGCGTTTGTAGGTGGTTTTATGGGTCTTTCTTCAAAGCGTAAAGCTTAAC
#> ASV2596  "AACGGAGAGGGCAAGTGTTATTCGTAATGACTGGGCGTAAGGCATACGTAGGTTGCTTTTTGAACTTTGTAACTAAAAACGATAAGGCGA
#> ASV13770 "AACGAGAGGAGCGAGTGTTATTCGTACTGATTGGGCGTAAAGGGTATGTAGGTTGTTAATCAAGTTTAAAGTTAAATACTGGGGCTCAAC
#> ASV13514 "TACGGAGGGGACGAGTGTTACTCGACATTATTGGGCGTAAAGGGTATGTAGGCGGTTTTCTAAGTTGAAAGTAAAAGACTAGAGCTTAAC
#> ASV12010 "AACGGGGGGATCAAGTGTTATTCAGAATCATTGGGCGTAAAGCGCATGTAGGCTGCCAATCAAGTTCATGATATCAATCCCTGCGCCCAA
#> ASV14108 "TACGGAAGAGGCAAACGTTACTCGGATTCACTGGGCGTAAAGGGTGAGTAGGCGGAATGGTAAGTCTAAGGTGAAAGACTGGGGCTCAAC
#> ASV11880 "TACGAAGGGGGCGAGTGTTCATCGGAATGACTGGGCGTGAAGGGTGTCTAGGCGGAGGTGCACTCATCGGGTTAAAGCCCTGGAAGAATC
#> ASV14053 "CACGGGGGATGCAAGCGTTACCCGGAATTATTGGGCGTAAAGAGTCTGTAGGCGGCAAAATACAGTAAGTTTGTCGTTAAATTTTGAAGC
#> ASV11933 "TACGGGGGATGCGAGCGTTATTCGGAATTACTGGGCGTAAAGCGTCTGTAGGTGATGTGATGTGTTCATTGTGACCGAAAAGGGCTTAAC
#> ASV12150 "TACGTGGGACGCGAGCGTTATCCGGAATGATTGGGCGTAAAGTGTCTGTAGGAGGTTTGGAAAGTTATTTGTTAAATTCCAAGGCACAAC
#> ASV13298 "GACGGAGGGAGCAAGCGTTGTCCGGAATTATTGGGCGTAAAGGGTCCGTAGGCGGCTCGCTAAGTCTATTGTGAAAGCCCAGGGCTTAAC
#> ASV13728 "TACGGAGGGTGCAAGCGTTGTCCGGATTTATTGGGTTTAAAGGGTGCGTAGGTGGTTTATTAAGTCAGTGGCGAAGGCGACACTCTGGCT
as.data.frame(tax_info)
#>           Kingdom          Phylum               Class           Order
#> ASV11922 Bacteria Patescibacteria       Parcubacteria            <NA>
#> ASV14121 Bacteria            <NA>                <NA>            <NA>
#> ASV13576 Bacteria            <NA>                <NA>            <NA>
#> ASV14412 Bacteria    Bacteroidota         Bacteroidia    Cytophagales
#> ASV14016 Bacteria  Pseudomonadota                <NA>            <NA>
#> ASV13972  Archaea   Nanoarchaeota        Nanoarchaeia Woesearchaeales
#> ASV3504  Bacteria  Pseudomonadota                <NA>            <NA>
#> ASV9890  Bacteria            <NA>                <NA>            <NA>
#> ASV2596  Bacteria            <NA>                <NA>            <NA>
#> ASV13770 Bacteria  Pseudomonadota Alphaproteobacteria            <NA>
#> ASV13514 Bacteria  Pseudomonadota Alphaproteobacteria   Rickettsiales
#> ASV12010 Bacteria            <NA>                <NA>            <NA>
#> ASV14108 Bacteria   Spirochaetota        Brachyspirae  Brachyspirales
#> ASV11880 Bacteria            <NA>                <NA>            <NA>
#> ASV14053 Bacteria Cyanobacteriota      Cyanobacteriia            <NA>
#> ASV11933 Bacteria            <NA>                <NA>            <NA>
#> ASV12150 Bacteria            <NA>                <NA>            <NA>
#> ASV13298 Bacteria            <NA>                <NA>            <NA>
#> ASV13728 Bacteria  Pseudomonadota Gammaproteobacteria Pseudomonadales
#>                       Family           Genus Species Subspecies       ASV
#> ASV11922                <NA>            <NA>    <NA>       <NA> ASV_11922
```

```
#> ASV14121                <NA>             <NA>    <NA>       <NA> ASV_14121
#> ASV13576                <NA>             <NA>    <NA>       <NA> ASV_13576
#> ASV14412   Cyclobacteriaceae      Roseivirga    <NA>       <NA> ASV_14412
#> ASV14016                <NA>             <NA>    <NA>       <NA> ASV_14016
#> ASV13972      Incertae Sedis SCGC AAA286-E23    <NA>       <NA> ASV_13972
#> ASV3504                 <NA>             <NA>    <NA>       <NA>  ASV_3504
#> ASV9890                 <NA>             <NA>    <NA>       <NA>  ASV_9890
#> ASV2596                 <NA>             <NA>    <NA>       <NA>  ASV_2596
#> ASV13770                <NA>             <NA>    <NA>       <NA> ASV_13770
#> ASV13514                <NA>             <NA>    <NA>       <NA> ASV_13514
#> ASV12010                <NA>             <NA>    <NA>       <NA> ASV_12010
#> ASV14108     Brachyspiraceae     Brachyspira    <NA>       <NA> ASV_14108
#> ASV11880                <NA>             <NA>    <NA>       <NA> ASV_11880
#> ASV14053                <NA>             <NA>    <NA>       <NA> ASV_14053
#> ASV11933                <NA>             <NA>    <NA>       <NA> ASV_11933
#> ASV12150                <NA>             <NA>    <NA>       <NA> ASV_12150
#> ASV13298                <NA>             <NA>    <NA>       <NA> ASV_13298
#> ASV13728 Endozoicomonadaceae  Endozoicomonas    <NA>       <NA> ASV_13728
#>
#> ASV11922  TACGGAGACCTCAAGCGTTATCCGGAATCATTGGGCGTAAAGCGTACCGATCGGTGGTTTGGTAAGTCTGTTGTTAAAGACCGAAGCTCA
#> ASV14121  TACGTAGGTGGCAAGCGTTATTCGGAATTACTGGGCGTAAAGCGTTCGTAGGCGATTTTGTAGGTCCTTGATTAAATACTTCAGCTCAAC
#> ASV13576  TACGTAGGGCGCGAGCGTTATCCGGATTTACTGGGCGTAAAGAGTCGTGTAGGTGGTTTGTTACATCATGCTTGAAATCCTATGGCTCAA
#> ASV14412  AACGTAGGACGCAAGCGTTATCCGGAATTACTGGGCGTAAAGCGCGCGCAGGCGGGGTGTTAAGTCAGCCGTGAAAGCTCCCGGCTAAAC
#> ASV14016 TACATGAGTGGTGAGCGTTGCTTTTAATTGTTAGGTGTAAAACATCCTGAGAAACAATTATGGGTTATATATGTTAGTATACTATAATTGG
#> ASV13972  TACCGGCGCCACAAGTGGAAATCACGATTATTGGGTCTAAAGTGTTCGTAGCTGGTTATATAAGTTTCTTGTTAAAGTTCTGTAGCTTAA
#> ASV3504  CACGATTAACCCAAGTCAATAGAAGCCGGCGTAAAGAGTGTTTTAGATCACCCCCTCCCCAATAAAGCTAAAACTCACCTGAGTTGTAAAA
#> ASV9890  GACGGGGGATGCGAGCGTTATCCGGAATGACTGGGCGTAAAGCGTTTGTAGGTGGTTTTATGGGTCTTTCTTCAAAGCGTAAAGCTTAACT
#> ASV2596   AACGGAGAGGGCAAGTGTTATTCGTAATGACTGGGCGTAAGGCATACGTAGGTTGCTTTTTGAACTTTGTAACTAAAAACGATAAGGCGA
#> ASV13770 AACGAGAGGAGCGAGTGTTATTCGTACTGATTGGGCGTAAAGGGTATGTAGGTTGTTAATCAAGTTTAAAGTTAAATACTGGGGCTCAACC
#> ASV13514  TACGGAGGGGACGAGTGTTACTCGACATTATTGGGCGTAAAGGGTATGTAGGCGGTTTTCTAAGTTGAAAGTAAAAGACTAGAGCTTAAC
#> ASV12010  AACGGGGGGATCAAGTGTTATTCAGAATCATTGGGCGTAAAGCGCATGTAGGCTGCCAATCAAGTTCATGATATCAATCCCTGCGCCCAA
#> ASV14108  TACGGAAGAGGCAAACGTTACTCGGATTCACTGGGCGTAAAGGGTGAGTAGGCGGAATGGTAAGTCTAAGGTGAAAGACTGGGGCTCAAC
#> ASV11880  TACGAAGGGGGCGAGTGTTCATCGGAATGACTGGGCGTGAAGGGTGTCTAGGCGGAGGTGCACTCATCGGGTTAAAGCCCTGGAAGAATC
#> ASV14053 CACGGGGGATGCAAGCGTTACCCGGAATTATTGGGCGTAAAGAGTCTGTAGGCGGCAAAATACAGTAAGTTTGTCGTTAAATTTTGAAGCC
#> ASV11933 TACGGGGGATGCGAGCGTTATTCGGAATTACTGGGCGTAAAGCGTCTGTAGGTGATGTGATGTGTTCATTGTGACCGAAAAGGGCTTAACT
#> ASV12150  TACGTGGGACGCGAGCGTTATCCGGAATGATTGGGCGTAAAGTGTCTGTAGGAGGTTTGGAAAGTTATTTGTTAAATTCCAAGGCACAAC
#> ASV13298  GACGGAGGGAGCAAGCGTTGTCCGGAATTATTGGGCGTAAAGGGTCCGTAGGCGGCTCGCTAAGTCTATTGTGAAAGCCCAGGGCTTAAC
#> ASV13728 TACGGAGGGTGCAAGCGTTGTCCGGATTTATTGGGTTTAAAGGGTGCGTAGGTGGTTTATTAAGTCAGTGGCGAAGGCGACACTCTGGCTT
```

```
unrooted_physeq_noEuk_clean <-
  unrooted_physeq_noEuk %>%
  pop_taxa(., long_branch_ids)

kingdom_tree_root <-
  ggtree(unrooted_physeq_noEuk_clean) +
  # color tips by kingdom
  geom_tippoint(mapping = aes(color = Kingdom)) +
  scale_color_manual(values = c("goldenrod1", "cornflowerblue", "grey")) +
  # Add the node label
  geom_text(aes(label=node), hjust= -0.5, vjust = -0.3, size = 2) +
  # Add title
  labs(title = "Unrooted Tree") +
  #move the legend to the bottom
```

```
  theme(legend.position = "bottom")
kingdom_tree_root
```



5.5 - Root the Tree —-

```
# Is the tree rooted?
is.rooted(phy_tree(unrooted_physeq_noEuk_clean))
#> [1] FALSE

# Let's midpoint root the tree
midpoint_rooted_tree <-
  phangorn::midpoint(
    reorder(phy_tree(unrooted_physeq_noEuk_clean), "cladewise")
  )

# Is the new tree rooted?
is.rooted(midpoint_rooted_tree)
#> [1] TRUE
```

5.6 - Create New Phyloseq Object —-

```
# a - Merge tree with the new unrooted_physeq_noEuk_clean ----
  physeq_midpointTree <-
    merge_phyloseq(unrooted_physeq_noEuk_clean, midpoint_rooted_tree)

  # Get vector of taxa names present in your merged OTU table
  valid_taxa <- taxa_names(physeq_midpointTree)

  # Prune the original tree to keep only these taxa
  pruned_tree <- keep.tip(midpoint_rooted_tree, valid_taxa)

  # Add the pruned, rooted tree back to the physeq object
  phy_tree(physeq_midpointTree) <- pruned_tree

  # Save both phyloseq objects with our tree object to one .RData file
  save(physeq_midpointTree,
           file = paste(dir_out_dat,
                        pref_out,
                        "physeq_midpointTree.RData", sep = ""))
```

Session Information

```
devtools::session_info()
#> - Session info ---------------------------------------------------------------
#>  setting  value
#>  version  R version 4.6.1 (2026-06-24 ucrt)
#>  os       Windows 11 x64 (build 26300)
#>  system   x86_64, mingw32
#>  ui       RTerm
#>  language (EN)
#>  collate  English_United States.utf8
#>  ctype    English_United States.utf8
#>  tz       America/New_York
#>  date     2026-10-27
#>  pandoc   3.6.3 @ C:/Program Files/RStudio/resources/app/bin/quarto/bin/tools/ (via rmarkdown)
#>  quarto   1.7.32 @ C:\\PROGRA~1\\RStudio\\RESOUR~1\\app\\bin\\quarto\\bin\\quarto.exe
#>
#> - Packages -------------------------------------------------------------------
#>  package              * version    date (UTC) lib source
#>  abind                  1.4-8      2024-09-12 [1] CRAN (R 4.6.1)
#>  ade4                   1.7-24     2026-03-21 [1] CRAN (R 4.6.1)
#>  ape                  * 5.8-1      2024-12-16 [1] CRAN (R 4.6.1)
#>  aplot                  0.3.2      2026-09-22 [1] CRAN (R 4.6.1)
#>  backports              1.5.1      2026-04-03 [1] CRAN (R 4.6.1)
#>  Biobase                2.72.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  BiocGenerics           0.58.1     2026-05-14 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  biomformat             1.40.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  Biostrings             2.80.2     2026-09-02 [1] Bioconductor 3.23 (R 4.6.1)
#>  cachem                 1.1.0      2024-05-16 [1] CRAN (R 4.6.1)
```

```
#>  checkmate              2.3.4      2026-02-03 [1] CRAN (R 4.6.1)
#>  cli                    3.6.6      2026-04-09 [1] CRAN (R 4.6.1)
#>  cluster                2.1.8.2    2026-02-05 [2] CRAN (R 4.6.1)
#>  clusterGeneration      1.3.8      2023-08-16 [1] CRAN (R 4.6.1)
#>  coda                   0.19-4.1   2024-01-31 [1] CRAN (R 4.6.1)
#>  codetools              0.2-20     2024-03-31 [2] CRAN (R 4.6.1)
#>  combinat               0.0-9      2026-09-14 [1] CRAN (R 4.6.1)
#>  crayon                 1.5.3      2024-06-20 [1] CRAN (R 4.6.1)
#>  data.table             1.18.6.1   2026-08-24 [1] CRAN (R 4.6.1)
#>  DBI                    1.3.0      2026-02-25 [1] CRAN (R 4.6.1)
#>  DECIPHER               3.8.1      2026-07-30 [1] Bioconductor 3.23 (R 4.6.1)
#>  DelayedArray           0.38.2     2026-05-26 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  DEoptim                2.2-8      2022-11-11 [1] CRAN (R 4.6.1)
#>  devtools               2.5.2      2026-04-30 [1] CRAN (R 4.6.1)
#>  dichromat              2.0-1      2026-07-22 [1] CRAN (R 4.6.1)
#>  digest                 0.6.39     2025-11-19 [1] CRAN (R 4.6.1)
#>  doParallel             1.0.17     2022-02-07 [1] CRAN (R 4.6.1)
#>  dplyr                * 1.2.1      2026-04-03 [1] CRAN (R 4.6.1)
#>  ellipsis               0.3.3      2026-04-04 [1] CRAN (R 4.6.1)
#>  evaluate               1.0.5      2025-08-27 [1] CRAN (R 4.6.1)
#>  expm                   1.0-1      2026-08-23 [1] CRAN (R 4.6.1)
#>  expss                * 0.11.7     2025-10-15 [1] CRAN (R 4.6.1)
#>  farver                 2.1.2      2024-05-13 [1] CRAN (R 4.6.1)
#>  fastmap                1.2.0      2024-05-15 [1] CRAN (R 4.6.1)
#>  fastmatch              1.1-8      2026-01-17 [1] CRAN (R 4.6.1)
#>  fontBitstreamVera      0.1.1      2017-02-01 [1] CRAN (R 4.6.1)
#>  fontLiberation         0.1.0      2016-10-15 [1] CRAN (R 4.6.1)
#>  fontquiver             0.2.1      2017-02-01 [1] CRAN (R 4.6.1)
#>  forcats              * 1.0.1      2025-09-25 [1] CRAN (R 4.6.1)
#>  foreach                1.5.2      2022-02-02 [1] CRAN (R 4.6.1)
#>  fs                     2.1.0      2026-04-18 [1] CRAN (R 4.6.1)
#>  gdtools                0.5.2      2026-10-15 [1] CRAN (R 4.6.1)
#>  generics               0.1.4      2025-05-09 [1] CRAN (R 4.6.1)
#>  GenomicRanges          1.64.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  ggfun                  0.2.2      2026-10-09 [1] CRAN (R 4.6.1)
#>  ggiraph                0.9.6      2026-02-21 [1] CRAN (R 4.6.1)
#>  ggplot2              * 4.0.3      2026-04-22 [1] CRAN (R 4.6.1)
#>  ggplotify              0.1.4      2026-10-12 [1] CRAN (R 4.6.1)
#>  ggtree               * 4.2.0      2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  glue                   1.8.1      2026-04-17 [1] CRAN (R 4.6.1)
#>  gridGraphics           0.5-1      2020-12-13 [1] CRAN (R 4.6.1)
#>  gtable                 0.3.6      2024-10-25 [1] CRAN (R 4.6.1)
#>  hms                    1.1.4      2025-10-17 [1] CRAN (R 4.6.1)
#>  htmlTable              2.5.0      2026-04-22 [1] CRAN (R 4.6.1)
#>  htmltools              0.5.9      2025-12-04 [1] CRAN (R 4.6.1)
#>  htmlwidgets            1.6.4      2023-12-06 [1] CRAN (R 4.6.1)
#>  igraph                 2.3.3      2026-06-26 [1] CRAN (R 4.6.1)
#>  IRanges                2.46.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  iterators              1.0.14     2022-02-05 [1] CRAN (R 4.6.1)
#>  jsonlite               2.0.0      2025-03-27 [1] CRAN (R 4.6.1)
#>  knitr                  1.52       2026-09-06 [1] CRAN (R 4.6.1)
#>  labeling               0.4.3      2023-08-29 [1] CRAN (R 4.6.1)
#>  lattice                0.22-9     2026-02-09 [2] CRAN (R 4.6.1)
```

```
#>  lazyeval              0.2.3      2026-04-04 [1] CRAN (R 4.6.1)
#>  lifecycle             1.0.5      2026-01-08 [1] CRAN (R 4.6.1)
#>  lubridate           * 1.9.5      2026-02-04 [1] CRAN (R 4.6.1)
#>  maditr              * 0.8.7      2026-01-19 [1] CRAN (R 4.6.1)
#>  magrittr              2.0.5      2026-04-04 [1] CRAN (R 4.6.1)
#>  maps                * 3.4.4      2026-10-23 [1] CRAN (R 4.6.1)
#>  MASS                  7.3-65     2025-02-28 [2] CRAN (R 4.6.1)
#>  Matrix                1.7-5      2026-03-21 [2] CRAN (R 4.6.1)
#>  MatrixGenerics        1.24.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  matrixStats           1.5.0      2025-01-07 [1] CRAN (R 4.6.1)
#>  memoise               2.0.1      2021-11-26 [1] CRAN (R 4.6.1)
#>  mgcv                  1.9-4      2025-11-07 [2] CRAN (R 4.6.1)
#>  mnormt                2.1.2      2026-01-27 [1] CRAN (R 4.6.1)
#>  multtest              2.68.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  nlme                  3.1-169    2026-03-27 [2] CRAN (R 4.6.1)
#>  numDeriv              2016.8-1.1 2019-06-06 [1] CRAN (R 4.6.1)
#>  optimParallel         1.0-3      2026-07-30 [1] CRAN (R 4.6.1)
#>  otel                  0.2.0      2025-08-29 [1] CRAN (R 4.6.1)
#>  pacman                0.5.1      2019-03-11 [1] CRAN (R 4.6.1)
#>  patchwork             1.3.2      2025-08-25 [1] CRAN (R 4.6.1)
#>  permute               0.9-10     2026-02-06 [1] CRAN (R 4.6.1)
#>  phangorn              2.12.1     2024-09-17 [1] CRAN (R 4.6.1)
#>  phyloseq            * 1.56.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  phytools            * 2.5-2      2025-09-19 [1] CRAN (R 4.6.1)
#>  pillar                1.11.1     2025-09-17 [1] CRAN (R 4.6.1)
#>  pkgbuild              1.4.8      2025-05-26 [1] CRAN (R 4.6.1)
#>  pkgconfig             2.0.3      2019-09-22 [1] CRAN (R 4.6.1)
#>  pkgload               1.5.3      2026-06-15 [1] CRAN (R 4.6.1)
#>  plyr                  1.8.9      2023-10-02 [1] CRAN (R 4.6.1)
#>  purrr               * 1.2.2      2026-04-10 [1] CRAN (R 4.6.1)
#>  quadprog              1.5-8      2019-11-20 [1] CRAN (R 4.6.1)
#>  R6                    2.6.1      2025-02-15 [1] CRAN (R 4.6.1)
#>  rappdirs              0.3.4      2026-01-17 [1] CRAN (R 4.6.1)
#>  RColorBrewer          1.1-3      2022-04-03 [1] CRAN (R 4.6.1)
#>  Rcpp                  1.1.2      2026-07-05 [1] CRAN (R 4.6.1)
#>  readr               * 2.2.0      2026-02-19 [1] CRAN (R 4.6.1)
#>  reshape2              1.4.5      2025-11-12 [1] CRAN (R 4.6.1)
#>  rlang                 1.3.0      2026-07-05 [1] CRAN (R 4.6.1)
#>  rmarkdown             2.32       2026-09-01 [1] CRAN (R 4.6.1)
#>  rstudioapi            0.19.0     2026-06-11 [1] CRAN (R 4.6.1)
#>  S4Arrays              1.12.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  S4Vectors             0.50.2     2026-08-23 [1] Bioconductor 3.23 (R 4.6.1)
#>  S7                    0.2.2      2026-04-22 [1] CRAN (R 4.6.1)
#>  scales                1.4.0      2025-04-24 [1] CRAN (R 4.6.1)
#>  scatterplot3d         0.3-45     2026-02-23 [1] CRAN (R 4.6.1)
#>  Seqinfo               1.2.0      2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  sessioninfo           1.2.4      2026-06-04 [1] CRAN (R 4.6.1)
#>  SparseArray           1.12.2     2026-05-01 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  stringi               1.8.9      2026-08-04 [1] CRAN (R 4.6.1)
#>  stringr             * 1.6.0      2025-11-04 [1] CRAN (R 4.6.1)
#>  SummarizedExperiment  1.42.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  survival              3.8-6      2026-01-16 [2] CRAN (R 4.6.1)
#>  systemfonts           1.3.2      2026-03-05 [1] CRAN (R 4.6.1)
```

```
#>  tibble               * 3.3.1      2026-01-11 [1] CRAN (R 4.6.1)
#>  tidyr                * 1.3.2      2025-12-19 [1] CRAN (R 4.6.1)
#>  tidyselect             1.2.1      2024-03-11 [1] CRAN (R 4.6.1)
#>  tidytree               0.4.8      2026-07-02 [1] CRAN (R 4.6.1)
#>  tidyverse            * 2.0.0      2023-02-22 [1] CRAN (R 4.6.1)
#>  timechange             0.4.0      2026-01-29 [1] CRAN (R 4.6.1)
#>  treeio                 1.36.1     2026-04-30 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  tzdb                   0.5.0      2025-03-15 [1] CRAN (R 4.6.1)
#>  usethis                3.2.2      2026-09-10 [1] CRAN (R 4.6.1)
#>  vctrs                  0.7.3      2026-04-11 [1] CRAN (R 4.6.1)
#>  vegan                  2.7-6      2026-08-31 [1] CRAN (R 4.6.1)
#>  withr                  3.0.3      2026-06-19 [1] CRAN (R 4.6.1)
#>  xfun                   0.60       2026-07-09 [1] CRAN (R 4.6.1)
#>  XVector                0.52.0     2026-04-28 [1] https://bioc-release.r-universe.dev (R 4.6.0)
#>  yaml                   2.3.12     2025-12-10 [1] CRAN (R 4.6.1)
#>  yulab.utils            0.2.5      2026-08-29 [1] CRAN (R 4.6.1)
#>
#>  [1] C:/Users/uax75/AppData/Local/R/win-library/4.6
#>  [2] C:/Program Files/R/R-4.6.1/library
#>  * -- Packages attached to the search path.
#>
#> ------------------------------------------------------------------------------
```